UX Case Study | Product & Packaging Design

The use of disposable paper cups is quite common for the purpose of serving hot liquids like tea, coffee, and milk in places such as college canteens, offices, railway stations, and street-side tea stalls. Although they are very useful and economical, the design of these cups often leads to usability problems while handling hot liquids.

Project Goal

To redesign disposable paper cups to improve grip, reduce heat discomfort, and prevent accidental spills while drinking hot beverages.

Problem Statement

Users often face problems in holding the disposable paper cups containing hot liquids securely because of the heat transfer, lack of grip, and unstable design, which may cause discomfort and spilling of the liquid.

Why this Problem matters?

Disposable paper cups are used on a large scale for hot drinks like tea and coffee. When these cups become too hot to handle, it may cause discomfort to the user, and there are chances of spilling the liquid or even getting a burn. This not only affects the user while drinking but also causes a problem in a crowded place like a canteen or an office. It is an issue that needs to be addressed for the betterment of the users.

Observations

- Most users hold the cup by the top rim to prevent the heat from the cup body.
- Some users may wrap tissues or napkins around the cup to prevent heat discomfort.
- Users may hold the cup carefully while walking to prevent spilling the hot drink.
- When the drink is very hot, users may change their grip frequently to prevent burning their fingers.
- In crowded areas such as canteens or tea stalls, the chances of spillage are high while holding the cup.

User Pain Points

- **Heat Discomfort –** The cup becomes too hot to hold when filled with hot beverages.
- **Lack of Grip –** The smooth surface of the paper makes it difficult to hold the cup.
- **Risk of Spillage –** There is a high risk of spilling hot drinks like tea or coffee.
- **Uncomfortable Holding Experience –** It will not be comfortable to hold the cup without a tissue or a sleeve.
- **Difficult to Carry While Walking –** Holding the cup safely while moving is challenging.

Edible Sleeve Cup + Finger indent cup

Ideation

Based on the identified user pain points, several design ideas were explored to improve the usability and safety of disposable paper cups when holding hot beverages.

Final Solution

The final solution is a disposable cup made of paper, with an edible outer sleeve to enhance grip and insulate the hand when carrying hot drinks. The textured sleeve made of biscuits insulates the hand and offers a better grip compared to smooth-edged cups made of paper.

A small area for finger support is included to ensure a secure grasp. After use, the edible sleeve can be consumed or recycled, eliminating the waste associated with separate sleeves.

This design improves comfort, stability, and sustainability, while maintaining the disposability of hot beverage cups.

Design inspiration from: https://www.lavazza.in/